Sahaj Bhadja

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Summary

Software Engineer with an M.S. in Computer Science and Advanced NLP coursework, building **full-stack** Python and TypeScript web apps on **AWS**, **payment integrations**, data sync pipelines, unit testing, and AI-assisted coding.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built fault-tolerant **asynchronous job** processing with Python FastAPI workers, explicit job-state workflows, retries, and failure isolation, processing 1K+ jobs at 3x throughput without cascading failures under concurrent load.
- Designed **multi-tenant** relational schemas with row-level security for 500+ users, tuned indexed queries to cut dashboard latency from 1.2s to 350ms, and ran data integrations at 99%+ reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built Python and React **web app** features and API integrations with timeouts, retries, and fallbacks across OpenAI, Anthropic, and Qwen models, serving 10K+ daily requests for 150+ instructors on AWS.
- **Debugged** OCR timeouts and model failures reported by instructors, reproduced them, fixed root causes with fallback routing and batch processing, and cut feedback turnaround 60% across grading workflows.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Designed **REST API** endpoints with pagination, field filtering, and caching for clients on low-bandwidth networks, improving response times 30% and cutting payload sizes across large-scale integrations used by distributed clients.
- Wrote 20+ automated unit and integration test suites with PyTest and Mockito, run nightly through Jenkins and GitHub-triggered branches across Windows, Linux, and macOS in Agile sprints.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built **authorization** with JWT, role-based access, and **invoice and billing** services in a Python Django microservices platform, syncing 2M+ records across locations through real-time WebSocket updates for customer and lead workflows.
- Shipped through Jenkins CI/CD with Docker and staged releases, and debugged production issues with Sentry, Azure Monitor, and Kibana, cutting deployment time 50% and reaching 95%+ successful build validation.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built customer-facing Flask APIs for bookings, pricing, and coupons with configurable eligibility rules, keeping responses under 200ms across 50K+ records with Firebase real-time updates on Android, iOS, and web.
- Integrated multi-currency **payment** providers through **idempotent webhooks** for refunds and disputes, cutting manual handling about 70%, and implemented wallet-signature login that removed password storage for wallet users.

Skills

Languages: Python, TypeScript, JavaScript, Java, SQL, HTML, CSS
Web & Backend: React, Next.js, Node.js, Django, FastAPI, Flask, REST APIs, Webhooks, Idempotency
Data & Cloud: PostgreSQL, MySQL, Data Modeling, Data Reconciliation, AWS, Docker, CI/CD, GitHub Actions
AI & ML: LLM APIs, NLP, Embeddings, PyTorch, AI Coding Assistants, Prompt Engineering, OCR
Practices: Data Structures, Algorithms, System Design, Git, Unit Testing, PyTest, Debugging, Code Reviews, Sentry, Linux

Projects

Payroll Contribution Sync Service | *Python, Django, PostgreSQL, Celery, AWS*

- Designed a service that pulls each pay period's **payroll** files from multiple providers, maps them to a common schema, calculates 401(k) deferrals and employer match, and **reconciles** totals before funding.

Payroll Integration Health Monitor | *TypeScript, React, Node.js, PostgreSQL*

- Designed a web dashboard that tracks every **payroll integration** sync, flags late or mismatched contribution files, keeps an **audit** log of fixes, and alerts the owning engineer before deadlines.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*

- Built a **fintech** platform on Kubernetes with CI/CD, Plaid bank linking, a token-based firewall, and real-time updates, supporting 50,000+ users and 1M+ transactions with MongoDB and Next.js.